

NHANES Audiometry 2017–2020: Complete Exploratory Data Analysis

Fuzzy Audiometric Index project — dataset characterisation

2026-08-14

Table of contents

Data Source and Files	2
Data Dictionary (key variables)	2
Cleaning Pipeline	2
Sample Flow	3
Demographics	3
Threshold Distributions	4
PTA-4 and WHO Severity	5
The Borderline Problem	6
Audiogram Configuration	6
Inter-Aural Asymmetry	7
Inter-Frequency Correlations	8
Missing Data Patterns	9
Adult-Only (20 y) Subset — Decision Notes	9
Key Takeaways	10

Data Source and Files

- **Exam file:** P_AUX.xpt — NHANES 2017–2020 (pre-pandemic) audiometry examination + hearing questionnaire, **5,147 participants**.
- **Demographic file:** P_DEMO.xpt (2017–2020 pre-pandemic combined release) — linked by SEQN.
- **Frequencies tested:** 500, 1000, 2000, 3000, 4000, 6000, 8000 Hz, air-conduction, both ears (14 threshold variables).
- **Additional data in file:** tympanometry, otoscopy, acoustic-reflex, hearing-questionnaire items.

Protocol note (critical): per NHANES protocol, audiometry in this cycle was administered to **children/adolescents aged 6–19 y** and **adults aged 70 y**. The file contains **no participants aged 20–69**.

Data Dictionary (key variables)

Variable	Meaning
SEQN	Participant identifier
AUXU{500,1k,2k,3k,4k,6k,8k}{R,L}	Air-conduction thresholds (dB HL) per ear
AUXTMEP{R,L}, AUXTCOMR/L	Tympanometric pressure / compliance
AUATYMTR/L	Tympanogram type
AUXU1K2{R,L}	1 kHz retest (QC)
AUQ010	Hearing-aid use item (adult subgroup only)
AUQ540, AUQ580	3+ months loud-noise exposure items (adult subgroup only)
RIDAGEYR, RIAGENDR (P_DEMO)	Age (years), sex

Cleaning Pipeline

NHANES sentinel codes are stored as numeric values and **must be converted to missing** before any statistic:

Code	Meaning
666	No response at maximum output
777	Refused
888	Could not obtain test result

Code	Meaning
999	Not tested
SAS subnormal (<1e−70)	Implied missing

Valid thresholds were clipped to −10 to 120 dB HL. All statistics below use cleaned values.

Sample Flow

Stage	Participants	Ears
Raw file	5,147	—
With audiometry exam record	5,147	—
Exam status “complete”	3,775	—
Clean ears (all 7 freqs, no sentinels)	2,314	3,539
— of which adults 20 y (i.e. 70 y per protocol)	1,125	1,980

Demographics

Age (full file, n = 5,147 with audiometry): mean 30.7 y, median 15.0 y, range 6.0–80.0 y.

Age-band composition (protocol effect):

Age band	Participants	%
6–19	3,654.0	71.0
20–39	nan	nan
40–59	nan	nan
60–69	nan	nan
70–79	917.0	17.8
80+	576.0	11.2

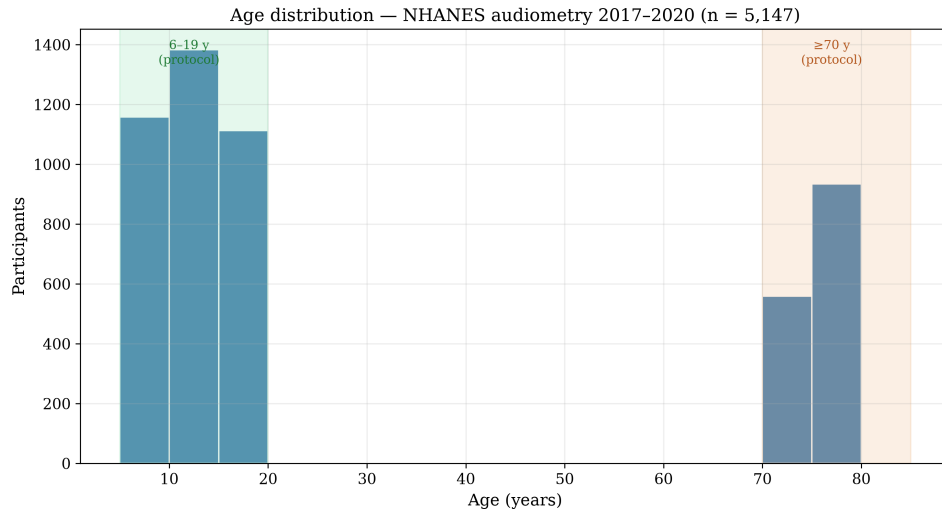


Figure 1: Age distribution

Sex: full file 48.8% female; clean-ear set 49.9%; adult clean set 51.4%.

Clean-ear set age: mean 47.8 y (SD 31.6), median 70.0 — the complete-threshold subset skews older because older adults who completed all frequencies (and children) both contribute, while the 6–19 group remains the largest single band.

Threshold Distributions

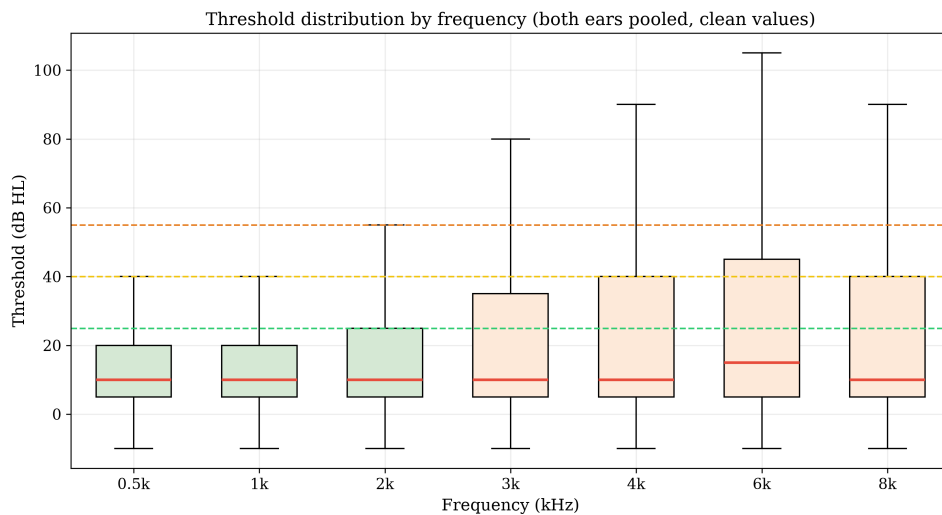


Figure 2: Threshold box plots by frequency

Frequency	n (pooled)	Mean (dB)	SD	Median
0.5k	7,284	14.3	13.5	10.0
1k	7,291	14.2	15.5	10.0
2k	7,373	18.1	19.7	10.0
3k	7,054	21.7	23.0	10.0
4k	7,349	22.5	25.3	10.0
6k	7,245	25.2	25.9	15.0
8k	7,140	23.3	27.3	10.0

Thresholds increase monotonically with frequency — the classic high-frequency loss pattern.

PTA-4 and WHO Severity

PTA-4 = mean of 500, 1000, 2000, 4000 Hz.

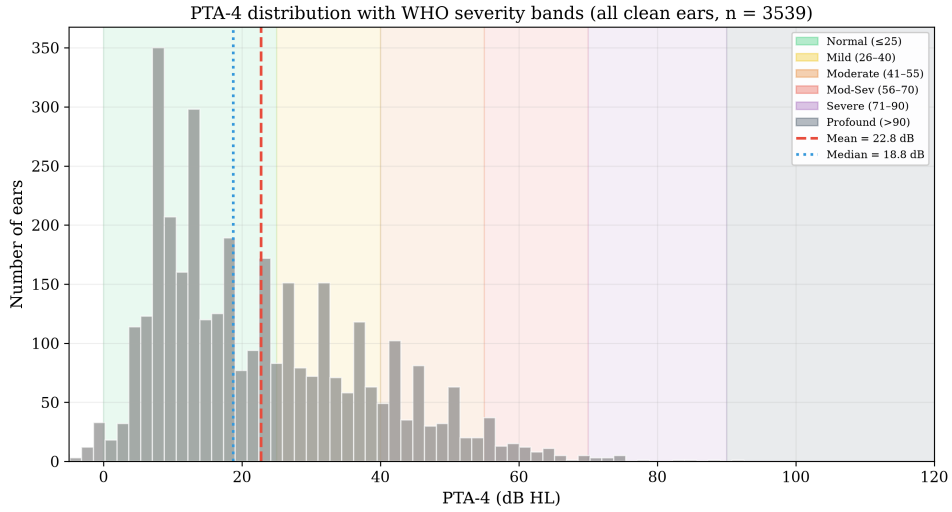


Figure 3: PTA-4 — all clean ears

Severity (WHO)	All clean ears	Adults 20 y
Normal	62.8%	37.5%
Mild	22.9%	38.3%
Moderate	11.4%	19.5%
Moderately Severe	2.4%	4.2%
Severe	0.4%	0.5%
Profound	0.0%	0.1%

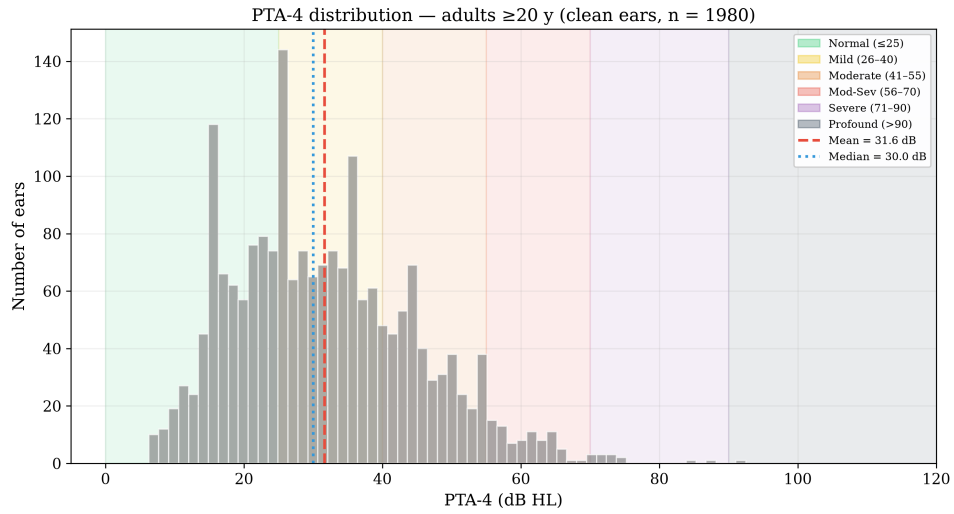


Figure 4: PTA-4 — adults ≥20 y

The Borderline Problem

Share of ears within ± 5 dB of a WHO severity boundary:

Boundary zone	dB range	All clean ears	Adults ≥20 y
Normal–Mild	20–30 dB	20.6%	32.0%
Mild–Moderate	35–45 dB	13.1%	22.2%
Moderate–Mod-Sev	50–60 dB	4.7%	8.2%
Mod-Sev–Severe	65–75 dB	0.7%	1.1%
Severe–Profound	85–95 dB	0.1%	0.2%

39.2% of all clean ears and **63.6%** of adult ears fall within ± 5 dB of a boundary — the population where crisp classification is least reliable.

Audiogram Configuration

Slope = threshold(4 kHz) – threshold(500 Hz).

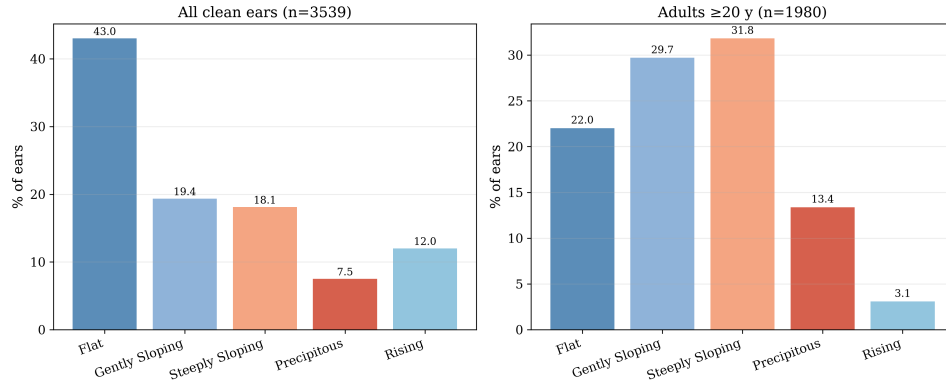


Figure 5: Configuration distribution

Configuration	All clean ears	Adults 20 y
Flat	43.0%	22.0%
Gently Sloping	19.4%	29.7%
Steeply Sloping	18.1%	31.8%
Precipitous	7.5%	13.4%
Rising	12.0%	3.1%

Inter-Aural Asymmetry

Bilateral participants only (n = 1,225); $|PTA-4 R - PTA-4 L|$:

- Symmetric (15 dB): 94.0%
- Mildly asymmetric (16–30 dB): 5.3%
- Moderately asymmetric (31–45 dB): 0.6%
- Severely asymmetric (>45 dB): 0.2%

Mean asymmetry 5.6 dB, median 3.8 dB.

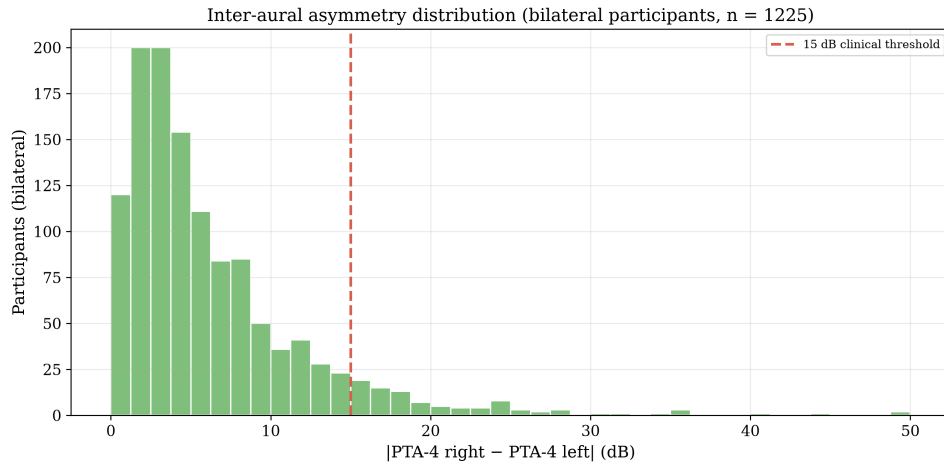


Figure 6: Inter-aural asymmetry

Inter-Frequency Correlations

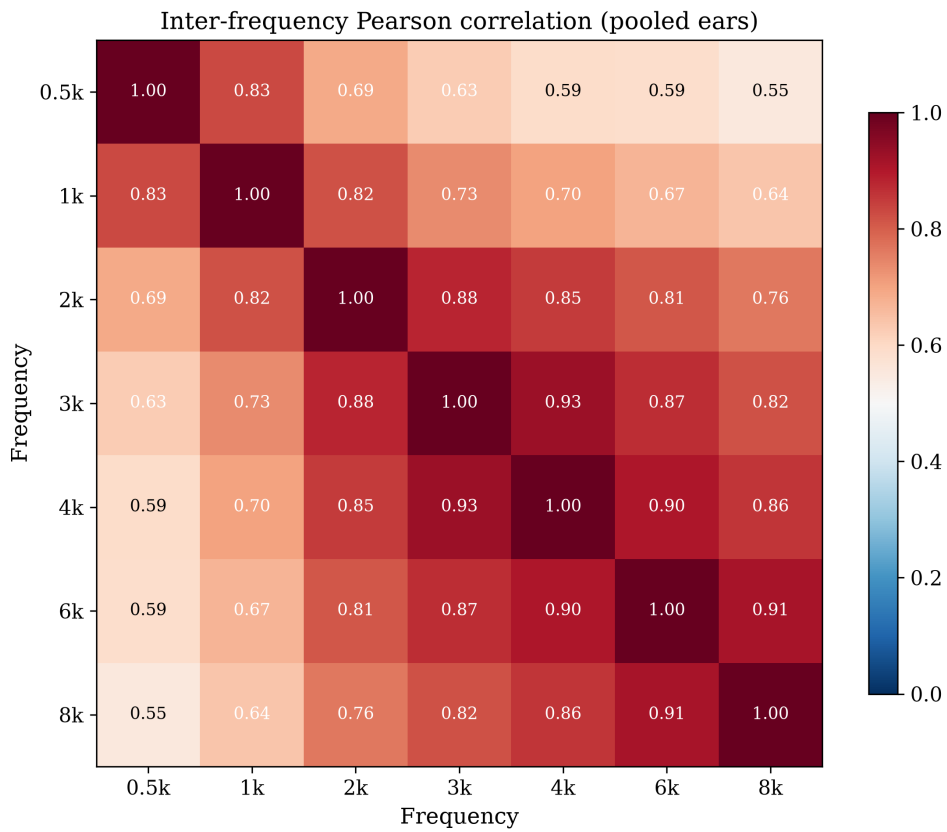


Figure 7: Correlation matrix

Adjacent-frequency correlations are high (mean $r = 0.88$); the 500 Hz–8 kHz correlation is moderate ($r = 0.55$) — frequency-specific information is partly independent of the PTA average.

Missing Data Patterns

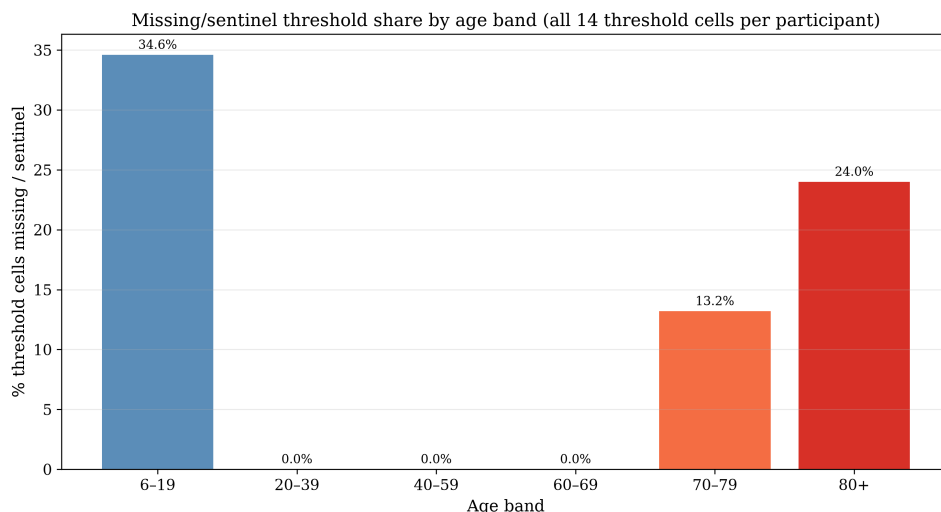


Figure 8: Missing/sentinel share by age band

Older participants have far higher missing/sentinel rates at high frequencies — the reason the clean-ear set (all 7 frequencies required) skews younger per participant even though adult ears dominate the ear count.

Adult-Only (20 y) Subset — Decision Notes

- Because the protocol tested only 6–19 and 70, “**adults 20**” is **exactly the 70 subgroup**: 1,125 participants, 1,980 clean ears.
- Adult clean-ear age: mean 75.7 (SD 3.8), median 76.0.
- Adult severity: Normal 37.5%; Mild 38.3%; Moderate 19.5%; Moderately Severe 4.2%; Severe 0.5%; Profound 0.1%.
- Borderline share: 63.6%; configuration: Flat 22.0%, Gently Sloping 29.7%, Steeply Sloping 31.8%, Precipitous 13.4%, Rising 3.1%.
- The FIS membership functions were optimised on the full training set (all ages); an adults-only analysis re-runs validation on this subset with the deployed parameters, or can re-optimize parameters on adult training data (larger change).

Key Takeaways

1. **The cohort is bimodal by design** (6–19 and 70); any “general adult population” framing is incorrect for this cycle.
2. **~39.2%** of ears sit within ± 5 dB of a severity boundary — the core motivation for graded fuzzy classification.
3. High-frequency thresholds are frequently missing in older adults; clean-ear selection (all 7 frequencies) is not age-neutral.
4. Configurations are predominantly flat/gently sloping in the full set; the adult subset shifts toward steeper slopes.
5. Inter-frequency correlations justify frequency-resolved (fuzzy) rather than PTA-only representation.